MIE312 Fluid Mechanics

Tutorial 11 - Flow Past Immersed Body

Howard Ho



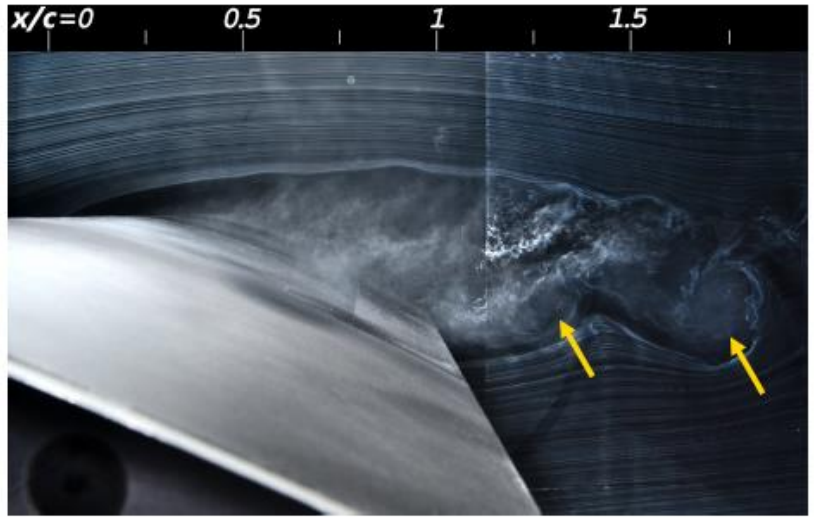
Learning Outcome

- Analysing problems in different types of external flow
- Understanding the concept of boundary layer
- Applying drag coefficient to find drag force on surfaces and bodies





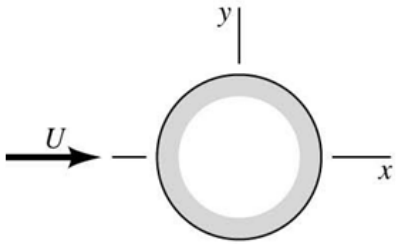


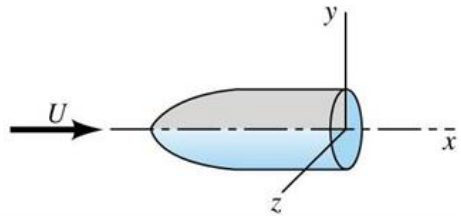


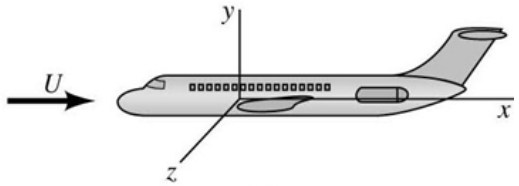
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External Flow Type

- Two-Dimensional: infinitely long and fixed cross-section (into the page)
- Axisymmetric: shaped by axis of symmetry
- Three-Dimensional: can still be symmetric (to a plane rather than an axis)



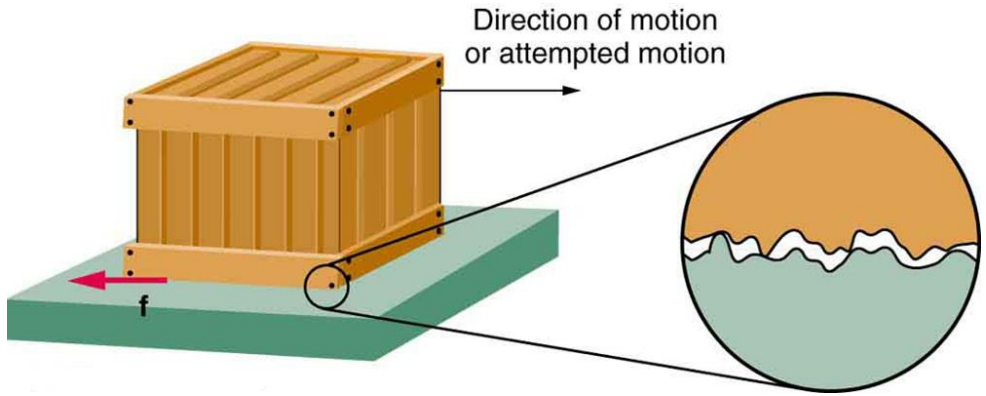


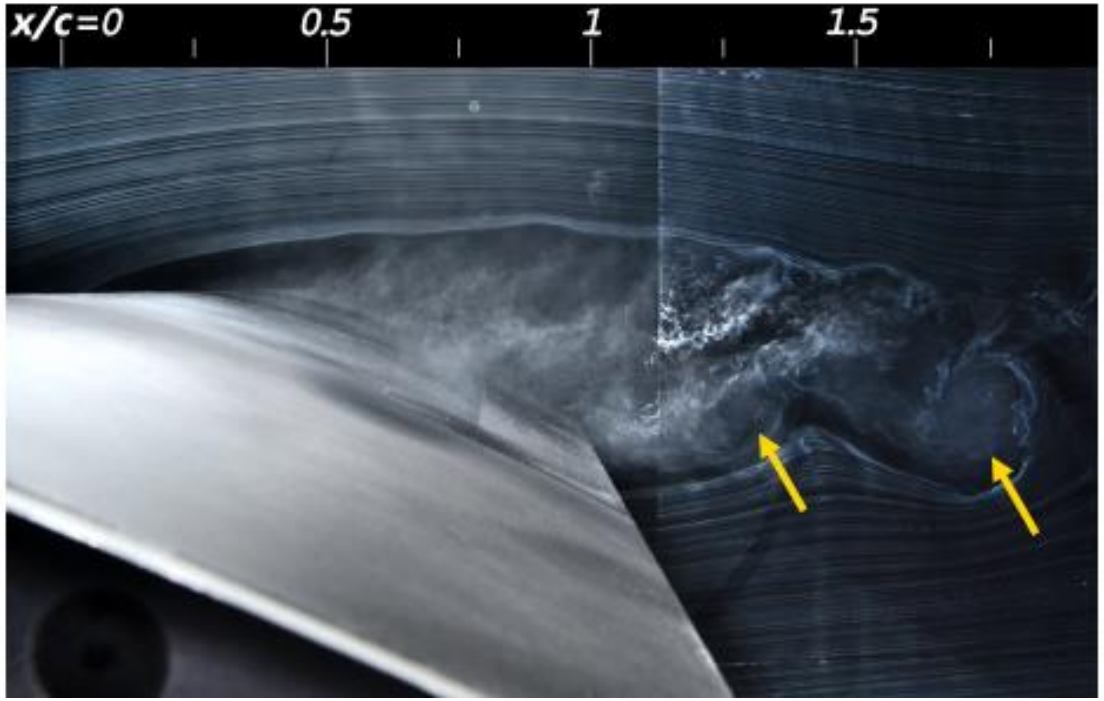


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Viscous and Inviscid Flow

- When the flow is inviscid, the shear (friction) is negligible, this often occurs further away from walls
  - Standard Bernoulli Equation can be used (if all requirements are met)
- When we examine flow close to walls or in a wake, significant shear occurs, and thus the flow is viscous
  - Such as pipe flow from Ch.6, Energy Transport Equation (RTT) is needed

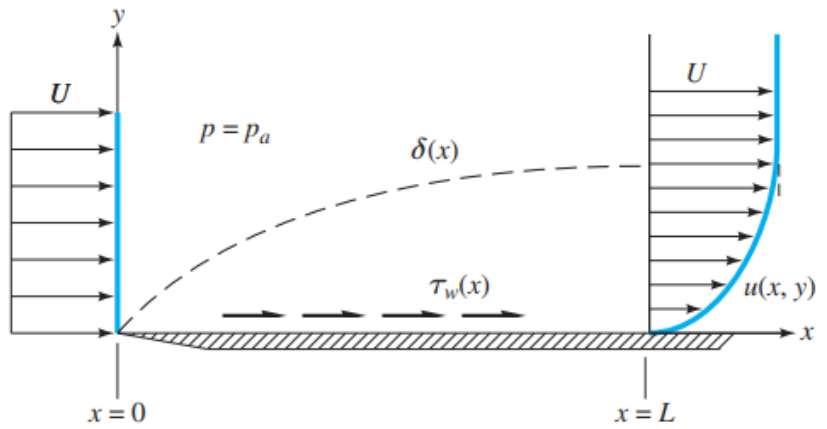




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Boundary Layer

- **Boundary layer (BL)** is a thin region of fluid near a solid surface where the effects of viscosity (fluid friction) are significant. The friction causes flow near the surface to slow down (to basically stationary if you look very close to the surface)

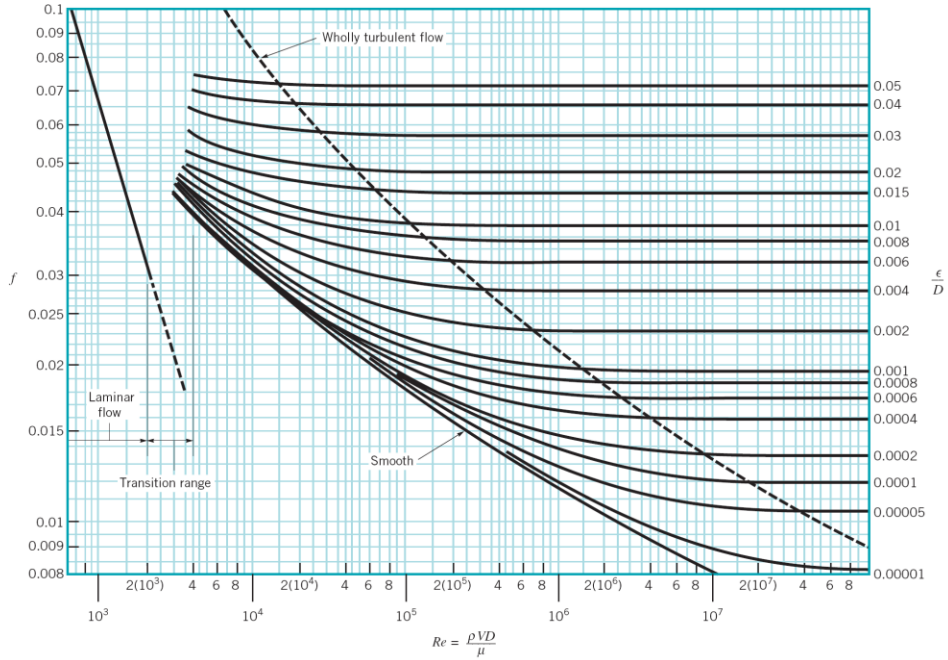


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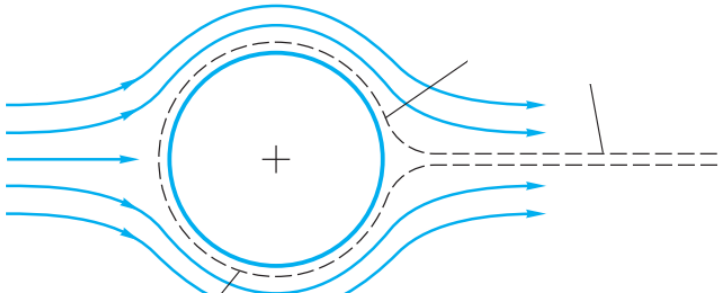
Reynolds Number (Flat Plate)

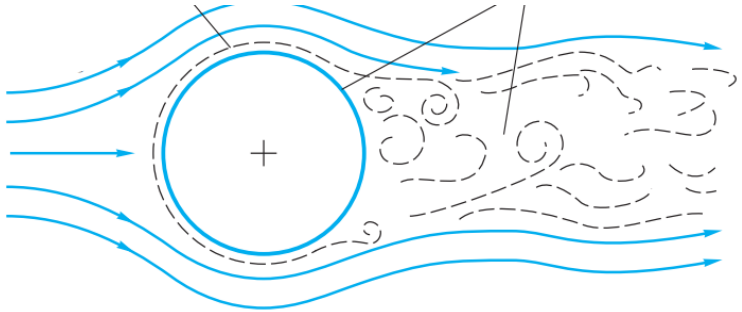
- Reynolds number for flow over flat plate can be calculated with

$$Re_L = \frac{\rho U L}{\mu}$$



- Laminar Re < 10^6
  - Orderly flow (clean, neat and reversible)
- Turbulence Re > 10^6
  - Chaotic (imagine chaos theory etc.)
- Technically, the transition is not a “light switch”
  - Transitional Region (treat that as a third flow regime)
  - Critical transition (assume it goes directly form laminar to turbulent at $\mathrm{Re}_{\mathrm{crit}}$)





Transitional flow is a beast in real life, we tend to avoid or minimize it for most applications



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Drag Force

- Drag Force F_D can be calculated as:

$$F_D = \frac{1}{2} C_D \rho V^2 A$$

where,

C_D = Drag coefficient (calculate or given in table/chart) []

ρ = Density of fluid $[kg/m^3]$

V = Free-stream (in-coming) velocity $[m/s]$

A = Area $[m^2]$

Fun fact, the C_D equation is constructed based on Buckingham Pi Theorem (Ch.5):

$$\Pi_1 = f(\Pi_2)$$

$$C_D = f\ (Re)$$

$$\frac{2F_D}{\rho V^2 A} = f\left(\frac{\rho U L}{\mu}\right)$$



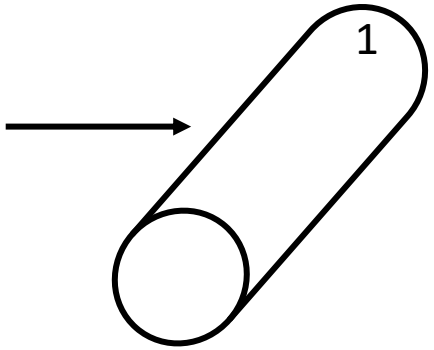


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Which Area to Pick?

$$F_D = \frac{1}{2} C_D \rho V^2 A$$

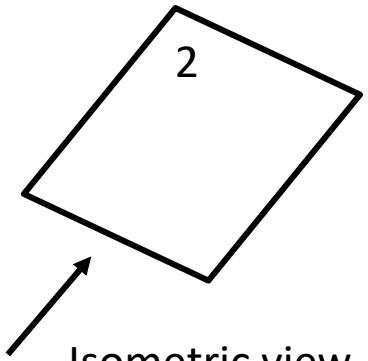
- The choice of *A* (Area) can be tricky, there are 3 main types:
  - 1. Frontal area: For thick, bluff bodies, such as spheres, cylinders.
  - 2. Planform area (Top-down view): For wide, flat bodies such as flat plates and airfoils.
  - 3. Wetted area: For surface ships and barges.



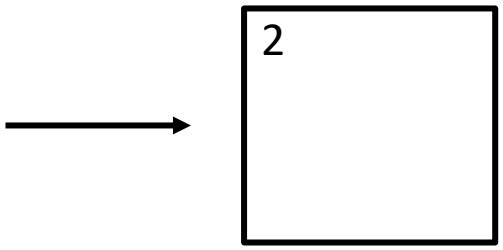
Isometric view



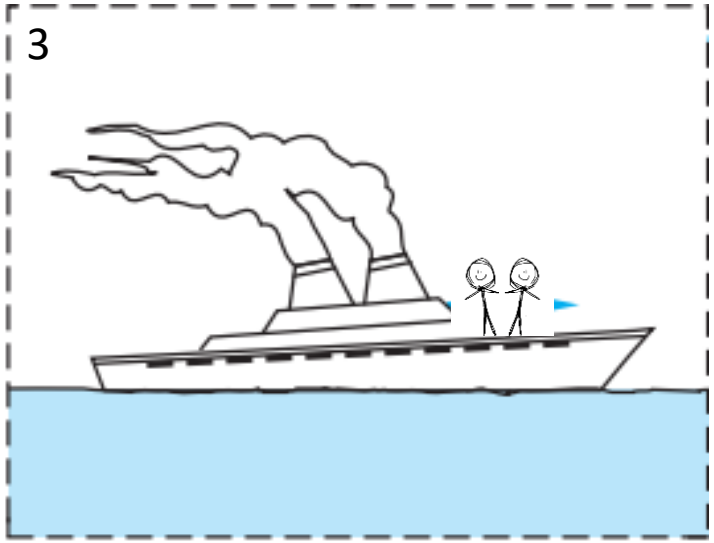
Frontal view



Isometric view



Top-down view



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Steps To Solve Drag Problems

1. Calculate Re, so we know if flow is laminar or turbulent ($Re = \frac{\rho UL}{\mu} = \frac{UL}{\nu}$)

2. Calculate or find C_D if not given (can also find BL height δ at this step)

   - Laminar Plate (Re < 10^6): $C_D = \frac{1.328}{Re^{1/2}}$; $\frac{\delta}{x} = \frac{5}{Re^{1/2}}$
   - Turbulent Plate (Re > 10^6): $C_D = \frac{0.031}{Re^{1/7}}$; $\frac{\delta}{x} = \frac{0.16}{Re^{1/7}}$
   - Transitional:



$$C_D \approx \begin{cases} \frac{0.031}{\mathrm{Re}_L^{1/7}} - \frac{1440}{\mathrm{Re}_L} & \mathrm{Re}_{trans} = 5 \times 10^5 \\ \frac{0.031}{\mathrm{Re}_L^{1/7}} - \frac{8700}{\mathrm{Re}_L} & \mathrm{Re}_{trans} = 3 \times 10^6 \end{cases}$$

3. Find the area type (frontal or top-down)

4. Calculate F_D with the drag force equation ($F_D = \frac{1}{2} C_D \rho V^2 A$) **X2 afterwards if needed**

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Question 1

7.26 Consider laminar flow past the square-plate arrangements in the figure below. Compared to the drag of a single plate (1), how much larger is the drag of four plates together as in configurations (a) and (b)? Explain your results.

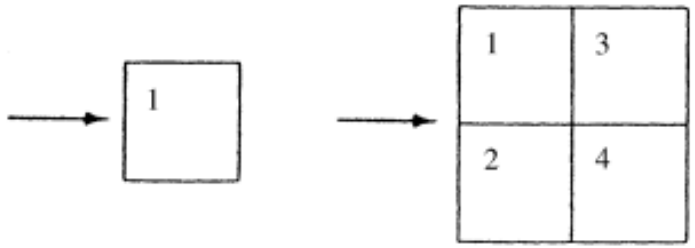


Fig. P7.26 (a)

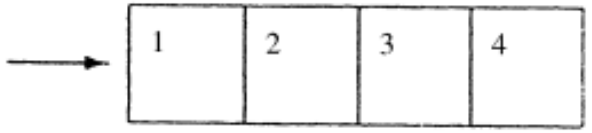
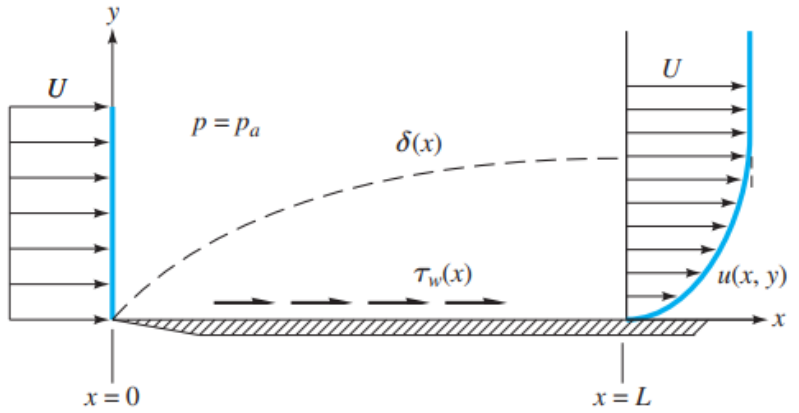
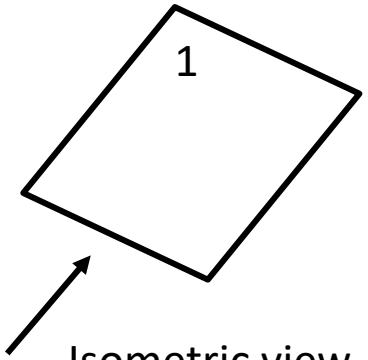


Fig. P7.26 (b)





Isometric view

4 Steps Guide
1) Find Re
2) Find C_D
3) Find A
4) Find F_D (x2 maybe?)

$$F_D = \frac{1}{2} C_D \rho V^2 A$$

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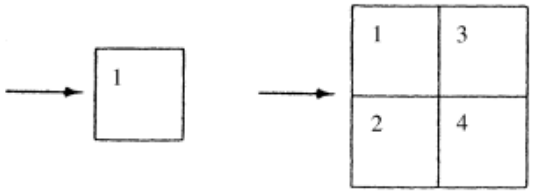


Fig. P7.26 (a)

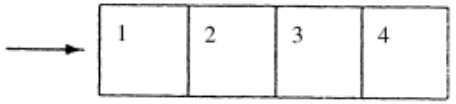


Fig. P7.26 (b)

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1) Find Re
2) Find C_D
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$$F_D = \frac{1}{2} C_D \rho V^2 A$$

8. BOUNDARY LAYERS & EXTERNAL

Params: $\delta^* = \int_0^\infty (1 - \frac{u}{U})dy; \quad \theta = \int_0^\infty \frac{u}{U}(1 - \frac{u}{U})dy$

Karman: $\frac{C_f}{2} = \frac{d\theta}{dx} + (H+2)\frac{\theta}{U}\frac{dU}{dx} \quad (H = \delta^*/\theta)$

Flat Plate:

Flow	δ/x	$C_{f,x}$	C_D
Lam.	$5/\sqrt{Re_x}$	$0.664/\sqrt{Re_x}$	$1.328/\sqrt{Re_L}$
Turb.	$0.16/Re_x^{1/7}$	$0.027/Re_x^{1/7}$	$0.031/Re_L^{1/7}$

Laminar displacement: $\delta^* = 1.7208x/\sqrt{Re_x} \approx 0.344\delta$

Separation: $\lambda = \frac{\theta^2}{\nu}\frac{dU}{dx}$; $\lambda_{sep} \approx -0.09$; $H_{sep} \approx 3.5$

Howarth: $x_{sep}/L \approx 0.12$; *Görtler:* $x_{sep}/L \approx 0.16$

Drag: $F_D = C_D A \frac{1}{2}\rho V^2$; **Sphere:** $C_D = 24/Re$ (Stokes)

Falling sphere: $U_t = \sqrt{\frac{4}{3}\frac{\rho_s - \rho}{\rho}\frac{gD}{C_D}}$; iterate with $Re = U_t D/\nu$.

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Question 2

A hydrofoil 1.2 ft long and 6 ft wide is placed in a seawater flow of 40 ft/s, with $\rho = 1.99$ slugs/ft^3 and $\nu = 0.000011$ ft^2/s. (a) Estimate the boundary layer thickness at the end of the plate. Estimate the friction drag for (b) turbulent smooth-wall flow from the leading edge, (c) laminar turbulent flow with $\text{Re}_{\text{trans}} = 5 \times 10^5$



4 Steps Guide

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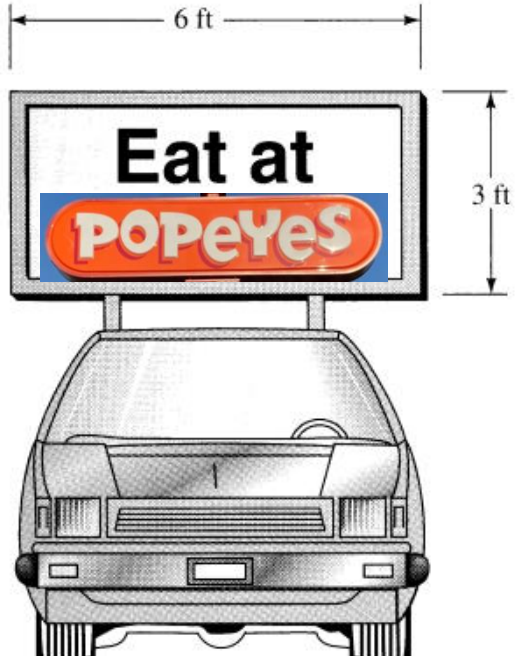
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Question 3

A Popeyes Pickup truck has a clean drag-area $C_D A$ of 35 ft^2. Estimate the horsepower required to drive the truck at 55 mi/h (a) clean and (b) with the 3-by 6-ft sign installed if the rolling resistance is 150 lbf at sea level. Assume the drag coefficient for the sign $C_D = 1.19$.



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